TARGET

1.Import the dataset and do usual exploratory analysis steps like checking the structure & characteristics of the dataset:

A. Data type of all columns in the "customers" table.

Field name	Type
customer_id	STRING
customer_unique_id	STRING
customer_zip_code_prefix	INTEGER
customer_city	STRING
customer_state	STRING

B. Get the time range between which the orders were placed.

QUERY:

```
SELECT min(order_purchase_timestamp) as Minimum_time_range,
 max(order_purchase_timestamp) as Maximum_time_range
FROM `Target.orders`
```

OUTPUT:

Job information | Results | Visualization | JSON | Execution de

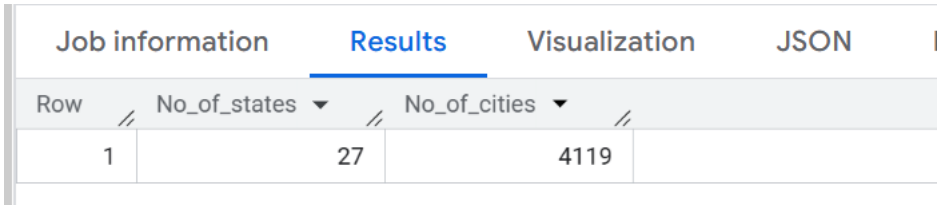
Row	Minimum_time_range	Maximum_time_range
1	2016-09-04 21:15:19 UTC	2018-10-17 17:30:18 UTC

C. Count the Cities & States of customers who ordered during the given period.

QUERY:

```
SELECT COUNT(DISTINCT c.customer_state)as No_of_states,
COUNT(DISTINCT c.customer_city)as No_of_cities
 FROM `Target.customers` c JOIN `Target.orders` o
  ON c.customer_id=o.customer_id
```

OUTPUT:



Job information | Results | Visualization | JSON

Row	No_of_states	No_of_cities
1	27	4119

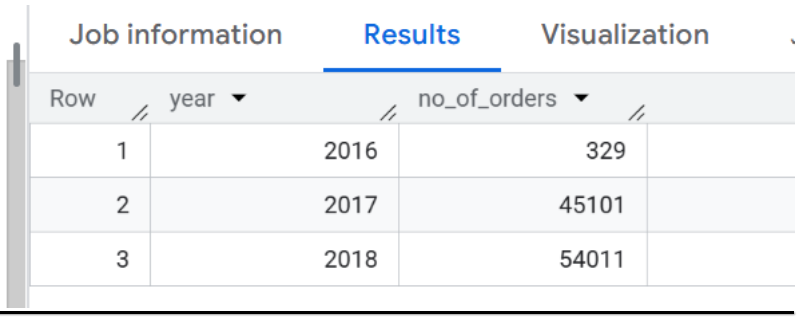
2.In-depth Exploration:

A.Is there a growing trend in the no. of orders placed over the past years?

QUERY:

```
SELECT
EXTRACT(YEAR FROM order_purchase_timestamp) AS year,
COUNT(order_id) AS no_of_orders
FROM `Target.orders`
GROUP BY year
ORDER BY year;
```

OUTPUT:

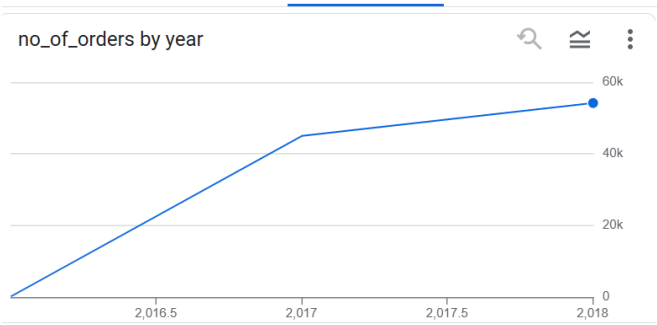


Job information | Results | Visualization

Row	year	no_of_orders
1	2016	329
2	2017	45101
3	2018	54011

Insights:

- Target is experiencing rapid growth in orders over the years.
- Orders increased from 329 from 2016 to 45101 in 2017 which is 13k%.



B. Can we see some kind of monthly seasonality in terms of the no. of orders being placed?

QUERY:

```
SELECT
EXTRACT(MONTH FROM order_purchase_timestamp) as month,
count(*)as no_of_orders
FROM `Target.orders`
GROUP BY month
ORDER BY month
```

OUTPUT

Row	month	no_of_orders
1	1	8069
2	2	8508
3	3	9893
4	4	9343
5	5	10573
6	6	9412
7	7	10318
8	8	10843
9	9	4305
10	10	4959

Insights & Recommendations



- Number of orders raised in mid of the year (i.e.) May, June, July, August comparatively other months.
- Discovered a significant decline in Month 9, indicating potential issues in customer retention or operations.
- Performed root cause analysis and recommended targeted marketing, retention strategies and operational improvements.
- Built a dashboard to monitor order trends and enable data-driven decision-making for sustained business growth.

C. During what time of the day, do the Brazilian customers mostly place their orders? (Dawn, Morning, Afternoon or Night)

QUERY:

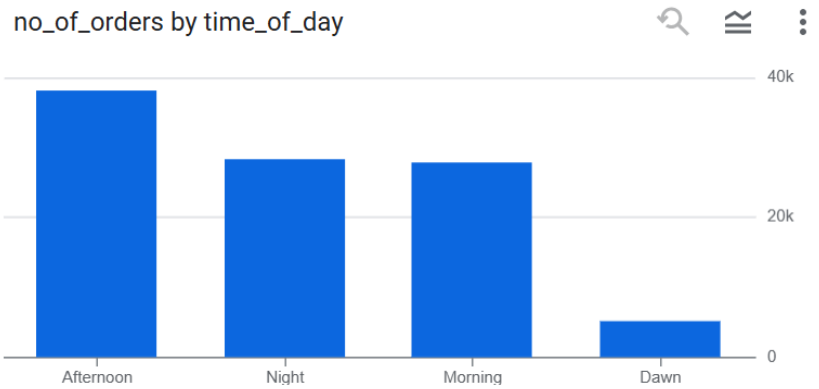
```
SELECT
  CASE
    WHEN EXTRACT(HOUR FROM order_purchase_timestamp) BETWEEN 0
AND 6 THEN 'Dawn'
    WHEN EXTRACT(HOUR FROM order_purchase_timestamp) BETWEEN 7
AND 12 THEN 'Morning'
    WHEN EXTRACT(HOUR FROM order_purchase_timestamp) BETWEEN 13
AND 18 THEN 'Afternoon'
  ELSE 'Night'
  END AS time_of_day,
```

```
    COUNT(order_id) AS no_of_orders
FROM `Target.orders`
GROUP BY time_of_day
ORDER BY no_of_orders desc
```

OUTPUT:

Row	time_of_day	no_of_orders
1	Afternoon	38135
2	Night	28331
3	Morning	27733
4	Dawn	5242

Insights:



Brazilian customers mostly placed their orders in the Afternoon. Customers are most active during the daytime/working hours.

Recommendations:

- Schedule push notification in the peak hours.
- Run ads/promotion in the afternoon hours.
- Ensure web/app performance is strong in the peak of afternoon.

3.Evolution of E-commerce orders in the Brazil region:

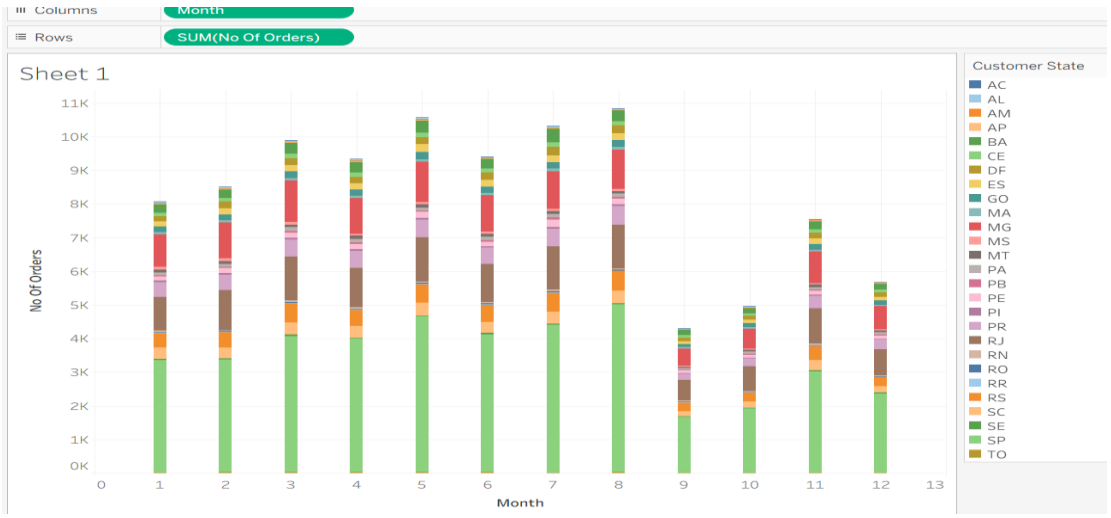
A. Get the month-ON-month no. of orders placed in each state.

QUERY:

```
SELECT
  EXTRACT(YEAR FROM order_purchase_timestamp) as year,
  EXTRACT(MONTH FROM order_purchase_timestamp) as month,
  count(*)as no_of_orders,
  c.customer_state
  FROM `Target.orders` o JOIN `Target.customers` c
  ON o.customer_id=c.customer_id
  GROUP BY year,month,customer_state
  ORDER BY year,month
```

OUTPUT:

Row	year	month	no_of_orders	customer_state
1	2016	9	1	RS
2	2016	9	1	RR
3	2016	9	2	SP
4	2016	10	40	MG
5	2016	10	113	SP
6	2016	10	8	CE
7	2016	10	56	RJ
8	2016	10	3	MT
9	2016	10	9	GO
10	2016	10	11	SC



Insights:

- Target started very slowing in the September 2016 with just 1-2 orders per state.
- SP was the biggest market from the beginning with 113 orders.
- RJ and MG also started quickly and showed strong demand.

Recommendations:

- Run targeted campaigns in low order states during peak months to grow their contribution.

B. How are the customers distributed across all the states?

QUERY:

```
SELECT DISTINCT customer_state,
    COUNT(customer_id)as no_of_customers
FROM `Target.customers`
GROUP BY customer_state
ORDER BY no_of_customers DESC
```

OUTPUT:

Row	customer_state	no_of_customers
1	SP	41746
2	RJ	12852
3	MG	11635
4	RS	5466
5	PR	5045
6	SC	3637
7	BA	3380
8	DF	2140
9	ES	2033
10	GO	2020

Insights & Recommendations:

- SP has the highest number of customers so far, it is Target Brazil largest and most important market.
- RJ and MG rows behind SP with potential customers and strong demand.
- Target should focus more on states with less customers and diversify its customer base.

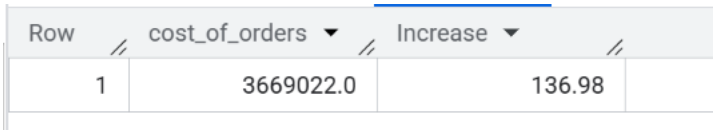
4.Impact on Economy: Analyse the money movement by e-commerce by looking at order prices, freight and others

A. Get the % increase in the cost of orders from year 2017 to 2018 (include months between Jan to Aug only).

QUERY:

```
 WITH yearly AS
 (
  SELECT
  t.cost_of_orders,
  ROUND((lead(t.cost_of_orders)over(order by year)-
t.cost_of_orders)/t.cost_of_orders*100,2)as Increase
FROM
  (SELECT
  EXTRACT(YEAR FROM o.order_purchase_timestamp)as year,
    ROUND(sum(p.payment_value))as cost_of_orders
  FROM`Target.orders` o JOIN `Target.payments` p
   ON o.order_id=p.order_id
   where EXTRACT(YEAR FROM o.order_purchase_timestamp)in(2017,2018) and
    EXTRACT(MONTH FROM o.order_purchase_timestamp) between 1 and 8
   GROUP BY year
  )t
)
SELECT *
FROM yearly
WHERE Increase is not null
```

OUTPUT:



Row	cost_of_orders	Increase
1	3669022.0	136.98

Insights:

- The total cost of orders grew significantly from 2017 to 2018(Jan-Aug) indicating strong business momentum.
- The percentage Increase shows that Target Corporation in Brazil was in the high growth phase during this period.

Recommendation:

- Check for product categories which drove 136% growth and double them in 2018.

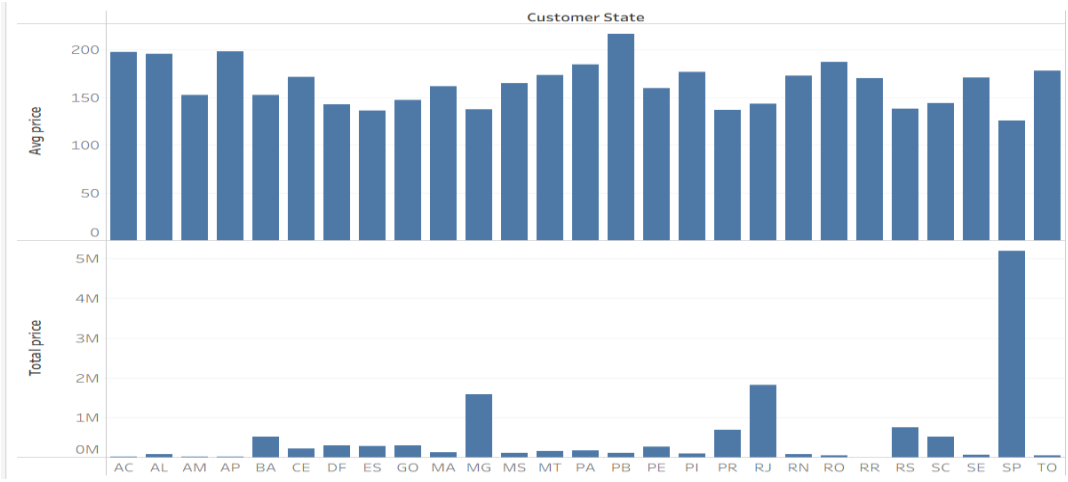
B. Calculate the Total & Average value of order price for each state

QUERY:

```
SELECT customer_state,
     ROUND(SUM(t.order_price),2)as Total_price,
     ROUND(AVG(t.order_price),2)as Avg_price
FROM (
   SELECT o.order_id,
        c.customer_state,
        SUM(ot.price)as order_price
   FROM `Target.order_items`ot JOIN `Target.orders` o
   ON ot.order_id=o.order_id
   JOIN `Target.customers` c
   ON o.customer_id=c.customer_id
   GROUP BY o.order_id,c.customer_state
   )  t
GROUP BY customer_state
ORDER BY Total_price DESC
```

OUTPUT:

Row	customer_state	Total_price	Avg_price
1	SP	5202955.05	125.75
2	RJ	1824092.67	142.93
3	MG	1585308.03	137.33
4	RS	750304.02	138.13
5	PR	683083.76	136.67
6	SC	520553.34	144.12
7	BA	511349.99	152.28
8	DF	302603.94	142.4
9	GO	294591.95	146.78
10	ES	275037.31	135.82



Insights &Recommendations:

- SP and RJ are typically the highest revenue states.
- Focus marketing spend on high-average price states to improve margins and investigate low average price with the high-volume states.

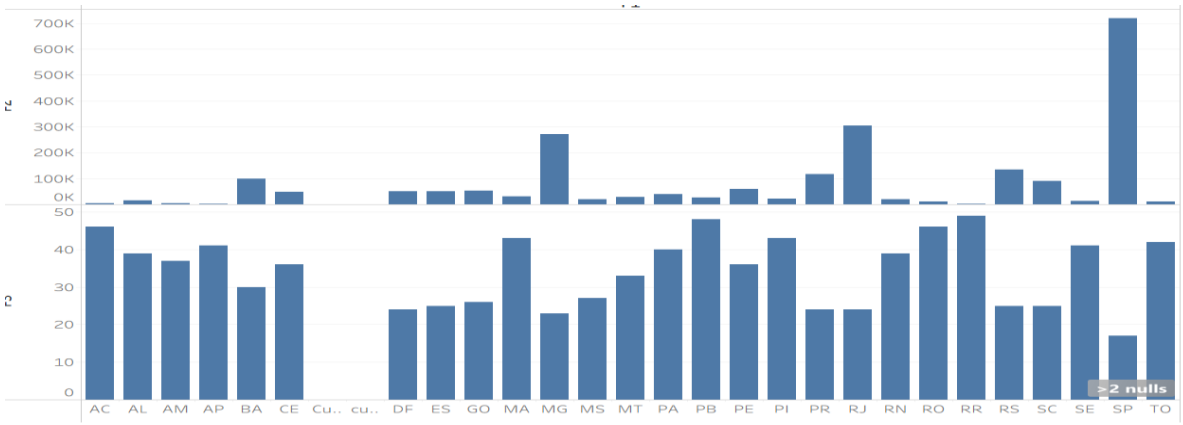
C. Calculate the Total & Average value of order freight for each state.

QUERY:

```
SELECT customer_state,
     ROUND(SUM(t.order_freight))as Total_price,
     ROUND(AVG(t.order_freight))as Avg_price
FROM (
   SELECT o.order_id,
        c.customer_state,
        SUM(ot.freight_value)as order_freight
   FROM `Target.order_items`ot JOIN `Target.orders` o
   ON ot.order_id=o.order_id
   JOIN `Target.customers` c
   ON o.customer_id=c.customer_id
   GROUP BY o.order_id,c.customer_state
   )  t
GROUP BY customer_state
```

OUTPUT:

Row	customer_state	Total_price	Avg_price
1	SP	718723.0	17.0
2	RJ	305589.0	24.0
3	MG	270853.0	23.0
4	RS	135523.0	25.0
5	PR	117852.0	24.0
6	BA	100157.0	30.0
7	SC	89660.0	25.0
8	PE	59450.0	36.0
9	GO	53115.0	26.0
10	DF	50626.0	24.0



Insights:

- SP has the highest revenue generating state.so it is the Brazil's core market for Target.
- MG holds third highest in revenue but lowest averages which means it drives orders through many smaller orders.
- PB has lowest revenue but highest averages which means small market but customers spend more there

Recommendation:

- States like SP and RJ needs to be priortise inventory,delivery speed and offers as they generate more revenue.
- PB and PA have the highest average value means customers spend more per order,increase brand visibility here to increase orders.
- MG has the highest orders but low average price so increase product bundles or recommendations to increase order value.

5.Analysis based on sales, freight and delivery time.

A. Find the no. of days taken to deliver each order from the order's purchase date as delivery time.
Also, calculate the difference (in days) between the estimated & actual delivery date of an order.

QUERY:

```
SELECT
  order_id,
 TIMESTAMP_DIFF(order_delivered_customer_date,
              order_purchase_timestamp,
              DAY) AS delivery_days,
 TIMESTAMP_DIFF(order_delivered_customer_date,
               order_estimated_delivery_date,
               DAY) AS diff_estimated_delivery
FROM `Target.orders`
WHERE order_delivered_customer_date IS NOT NULL and order_status='delivered'
```

OUTPUT:

w	order_id	delivery_days	diff_estimated_d...
1	bfbd0f9bdef84302105ad712db...	54	36
2	98974b076b01553d49ee64679...	43	-6
3	c4b41c36dd589e901f6879f25a...	36	-14
4	d2292ff2201e74c5db154d1b7a...	29	-20
5	95e01270fcbae986342340010...	30	-19
6	ed8c7b1b3eb256c70ce0c7423...	44	-5
7	5cc475c7c03290048eb2e742c...	68	18
8	6b3ee7697a02619a0ace2b3f0a...	47	-2
9	3b2ca3293a7ce539ea2379d70...	43	-7
10	b2f92b2f7047cd8b35580d629d...	43	-7

- Negative values (i.e. -6) shows that the ordered delivered 6 days early.
- 0 means order delivered on time.
- Positive values (i.e. 36) shows that the ordered delivered 36 days later than the estimated delivery.

Insights

- Late deliveries indicates that the systematic logistic gaps.
- Might experiencing challenges in remote or in low priority areas.

Recommendations

- Revisit estimated delivery date calculations to set more realistic expectations, reducing customer disappointments.
- Partnering with local logistics providers can reduce the delivery time.

B. Find out the top 5 states with the highest & lowest average freight value.

QUERY:

```
WITH state_avg AS (
  SELECT
    c.customer_state,
    AVG(ot.freight_value) AS avg_freight_value
  FROM `Target.order_items` ot
  JOIN `Target.orders` o
    ON ot.order_id = o.order_id
  JOIN `Target.customers` c
    ON o.customer_id = c.customer_id
  GROUP BY c.customer_state
),

top_5 AS (
  SELECT
    customer_state,
    DENSE_RANK()OVER (ORDER BY avg_freight_value DESC) AS rn
  FROM state_avg
),

bottom_5 AS (
  SELECT
    customer_state,
    DENSE_RANK()OVER (ORDER BY avg_freight_value ASC) AS rn
  FROM state_avg
)

SELECT
  t.customer_state AS highest_5_states,
  b.customer_state AS lowest_5_states
FROM top_5 t
JOIN bottom_5 b
  ON t.rn = b.rn
WHERE t.rn <= 5;
```

OUTPUT:

Row	highest_5_states	lowest_5_states
1	RR	SP
2	PB	PR
3	RO	MG
4	AC	RJ
5	PI	DF

Insights & Recommendations:

- SP, PR, MG, RJ, DF have the lowest average freight value, benefitting from dense delivery networks.
- RR, PB, RO, AC, PI have the highest average freight value, might not have enough delivery networks to reach customers.
- Expand warehouses presence closer to high freight values to shorten delivery time and bring freight cost down.

C. Find out the top 5 states with the highest & lowest average delivery time.

QUERY:

```
WITH state_delivery AS (
  SELECT
    c.customer_state,
    AVG(
      TIMESTAMP_DIFF(o.order_delivered_customer_date,
                     o.order_purchase_timestamp,
                     DAY)
    ) AS avg_delivery_time
  FROM `Target.orders` o
  JOIN `Target.customers` c
    ON o.customer_id = c.customer_id
  WHERE o.order_delivered_customer_date IS NOT NULL AND
o.order_status='delivered'
  GROUP BY c.customer_state
),
top_5 AS (
  SELECT
    customer_state,
    DENSE_RANK()OVER (ORDER BY avg_delivery_time DESC) AS rn
  FROM state_delivery
),

bottom_5 AS (
  SELECT
    customer_state,
```

```
   DENSE_RANK()OVER (ORDER BY avg_delivery_time ASC) AS rn
  FROM state_delivery
)

SELECT
  t.customer_state AS highest_delivery_time_states,
  b.customer_state AS lowest_delivery_time_states
FROM top_5 t
JOIN bottom_5 b
  ON t.rn = b.rn
WHERE t.rn <= 5;
```

OUTPUT:

Row	highest_delivery_time_states	lowest_delivery_time_states
1	RR	SP
2	AP	PR
3	AM	MG
4	AL	DF
5	PA	SC

Insights &Recommendations:

- RR, AP, AM, AL, PA are the states have highest average delivery time. Target need to work more on these states and provide local logistics for reducing delivery time.
- SP, PR, MG, DF, SC are the states which have lowest average delivery time.

D. Find out the top 5 states where the order delivery is really fast as compared to the estimated date of delivery.

QUERY:

```
WITH state_delivery_speed AS (
 SELECT
  c.customer_state,
  ROUND(AVG(
    TIMESTAMP_DIFF(o.order_delivered_customer_date,
                   o.order_estimated_delivery_date,
                   DAY)
  ),2)AS avg_delivery_diff
 FROM `Target.orders` o
 JOIN `Target.customers` c
  ON o.customer_id = c.customer_id
 WHERE o.order_delivered_customer_date IS NOT NULL
 GROUP BY c.customer_state
```

```
)
SELECT
  customer_state,
  avg_delivery_diff
FROM state_delivery_speed
ORDER BY avg_delivery_diff ASC
LIMIT 5;
```

OUTPUT:

Row	customer_state	avg_delivery_diff
1	AC	-19.76249999999...
2	RO	-19.13168724279...
3	AP	-18.73134328358...
4	AM	-18.60689655172...
5	RR	-16.41463414634...

Insights

- AC, RO, AP, AM, RR are the top 5 states where the order delivery is really fast.

Recommendations:

- AC, RO, AP, AM, RR are recommended to provide realistic delivery timing rather than inaccurate estimates.

6. Analysis based on the payments:

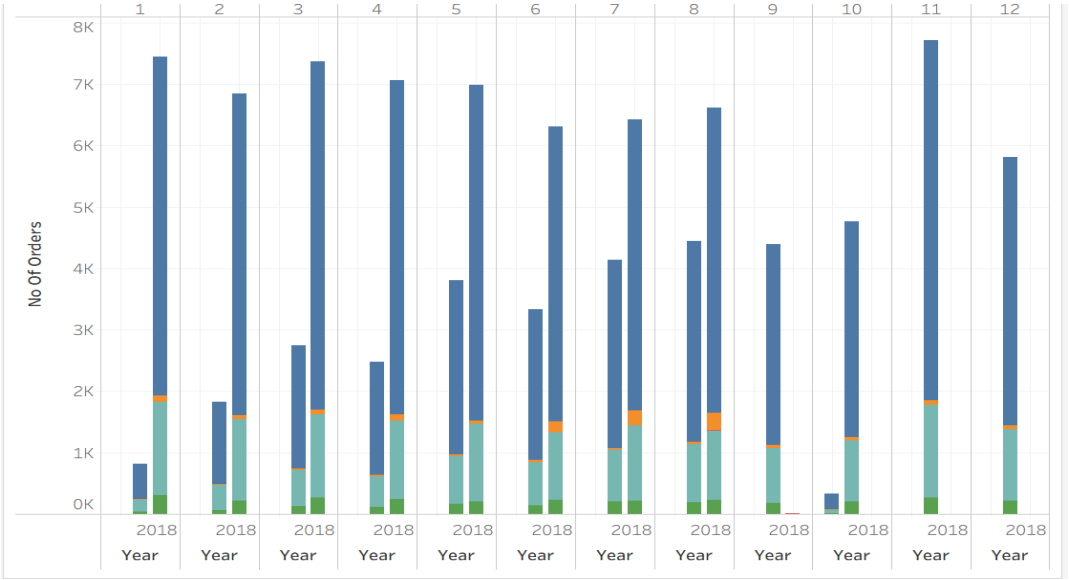
A. Find the month on month no. of orders placed using different payment types.

QUERY:

```
SELECT
    EXTRACT(YEAR FROM o.order_purchase_timestamp)AS year,
    EXTRACT(MONTH FROM o.order_purchase_timestamp)AS month,
    p.payment_type,
    COUNT(DISTINCT o.order_id)as no_of_orders
FROM `Target.orders` o join `Target.payments` p
ON o.order_id=p.order_id
GROUP BY year,month,p.payment_type
ORDER BY year,month,p.payment_type
```

OUTPUT:

Row	year	month	payment_type	no_of_orders
1	2016	9	credit_card	3
2	2016	10	UPI	63
3	2016	10	credit_card	253
4	2016	10	debit_card	2
5	2016	10	voucher	11
6	2016	12	credit_card	1
7	2017	1	UPI	197
8	2017	1	credit_card	582
9	2017	1	debit_card	9
10	2017	1	voucher	33



Insights&Recommendations:

- Credit card is typically the dominant payment type month-on-month, reflecting customer preference for deferred payment options in Brazil.
- Offer EMI or instalment incentives on credit card payments during peak months to further drive order volumes.

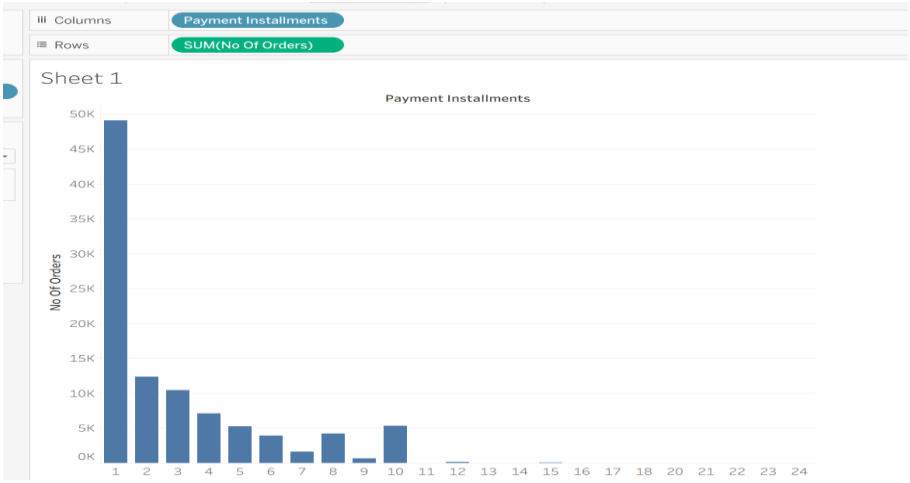
B. Find the no. of orders placed on the basis of the payment installments that have been paid.

QUERY:

```
SELECT
   payment_installments,
   COUNT(DISTINCT order_id) AS no_of_orders
FROM `Target.payments`
WHERE payment_installments >0
GROUP BY payment_installments
ORDER BY payment_installments;
```

OUTPUT:

Row	payment_installm...	no_of_orders
1	1	49060
2	2	12389
3	3	10443
4	4	7088
5	5	5234
6	6	3916
7	7	1623
8	8	4253
9	9	644
10	10	5315



Insights &Recommendations:

- Most customers (49,060) prefer paying in a single instalment, indicating a preference for full upfront payment.
- Orders decline steadily from instalment 2–7, but spike again at 8 and 10, suggesting high-value purchases being split into longer EMI plans.
- Promote 2–6 instalment options at checkout to encourage more customers to split payments and potentially increase order value.
- Introduce category-specific EMI plans for high-value products to explain the spike at instalments 8 and 10.

Business Observation:

- Target Brazil showed strong growth between 2016 – 2018, with SP, RJ and MG has the majority of orders and revenue.
- Afternoon hours and credit card payments dominate much on the customer behaviour.
- Logistics remain the key challenge focus on improving delivery efficiency.
- Expand the low revenue generating states will be crucial for sustainable revenue growth.